Manifold-Aware Concept Erasure and Evaluation

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Abstract

Concept erasure methods that operate in the ambient representation space can push representations off the data manifold, creating spurious correlations rather than removing information. We argue that both erasure and its evaluation should respect the intrinsic geometry of the representation manifold. On the erasure side, we introduce Tangent Gradient-based Concept Erasure (TanGCE), which constrains erasure gradients to the local tangent space estimated via neighborhood PCA. On the evaluation side, we adopt a k-nearest-neighbor mutual information estimator that measures concept information through local neighborhood structure—the same geometric lens that TanGCE uses for intervention. We complement this with unsupervised sparse autoencoder diagnostics that detect feature-level redistribution without requiring control-task labels. Across three concept domains—sycophancy, gender, and safety—in Qwen2.5-0.5B, and 40 facial attributes in CLIP ViT-B/32 representations of CelebA, TanGCE achieves the lowest nonlinear leakage while preserving control-task accuracy and maintaining near-zero feature redistribution. A λ sweep reveals that ambient erasure at high step sizes *increases* concept mutual information by laundering it into off-manifold displacement, a failure mode that TanGCE avoids by construction.

1 Introduction

Effective concept erasure in large language models requires a delicate balance between removing targeted information and preserving the model's broader utility. While gradient-based methods offer a path toward targeted intervention, they often suffer from substantial performance degradation. A natural hypothesis is that this collapse is driven in part by off-manifold edits: erasure directions computed in the high-dimensional ambient space can point orthogonal to the data manifold, displacing representations into low-density regions where the model's computation becomes unstable.

To address this failure mode, we propose Gradient-based Concept Erasure with Tangent projection (`gce_tangent`). By constraining erasure gradients to the local tangent space of the representation manifold, `gce_tangent` aims to remove concept-specific structure while preserving more of the underlying distributional geometry. This approach contrasts with `gce_ambient`, which applies raw gradients in the full hidden space and therefore provides a direct baseline for the value of tangent projection.

Our contributions are threefold:

1. **TanGCE: manifold-aware concept erasure.** We introduce Tangent Gradient-based Concept Erasure (TanGCE), which constrains erasure gradients to the local tangent space of the representation manifold. By retaining the tangent-projected gradient norm as a per-sample confidence weight, TanGCE naturally attenuates erasure for samples whose concept signal is predominantly off-manifold. We demonstrate that TanGCE achieves the lowest nonlinear leakage across three concept domains—sycophancy, gender, and safety—while preserving downstream task performance.

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2. **Manifold-aware mutual information evaluation.** We adopt a k-nearest-neighbor MI estimator that measures concept information through local neighborhood structure on the representation manifold. The estimator computes $I(\mathbf{X}; y)$ by examining, for each point, how many cross-class neighbors fall within the distance to its k-th same-class neighbor—a purely geometric quantity that captures all information (linear and nonlinear) without assuming a probe architecture. This creates a coherent geometric framework: TanGCE uses local PCA neighborhoods to estimate tangent spaces for erasure, while the k-NN MI estimator uses local ball neighborhoods to estimate information content. Both operate through local neighborhood structure rather than global ambient-space assumptions, and both degrade gracefully when the manifold assumption is only approximately satisfied. We compute MI for both concept labels (should decrease after erasure) and control labels (should remain unchanged), yielding a complete picture of erasure selectivity.

3. **Unsupervised SAE-based erasure diagnostics.** We propose three metrics computed via sparse autoencoders trained on a general-purpose corpus (no concept labels needed): *concept feature removal rate* (fraction of concept-correlated SAE features disrupted), *collateral damage rate* (fraction of non-concept features disrupted), and *feature redistribution score* (fraction of previously non-concept features that become concept-correlated after erasure). These metrics measure the surgicality and precision of erasure without requiring annotated control-task labels, and reveal failure modes invisible to downstream probes—including the critical finding that ambient erasure at high λ *creates* new concept correlations through off-manifold displacement (Section 5.4).

We evaluate on sycophancy as the primary nonlinear concept, gender (Bias in Bios) as a diagnostic baseline, and safety as a secondary case study in Qwen2.5-0.5B at layer 8, and on 40 binary facial attributes in CLIP ViT-B/32 representations of CelebA to demonstrate cross-modal generalization. Under a fair validation-based hyperparameter protocol—where all methods search the same λ grid and select by lowest validation nonlinear probe accuracy—TanGCE achieves the lowest nonlinear leakage across all domains while maintaining near-zero feature redistribution and preserving control-task accuracy. On CelebA, TanGCE wins 37 of 40 attribute erasure tasks, reducing mean nonlinear leakage to 0.707 (vs. 0.839 for LEACE), enabled by its ability to safely scale to λ^*=8.0 where ambient methods cannot.

2 Notation and Formal Framework

We define the notation used throughout, then give the formal definitions of TanGCE, the k-NN MI estimator, and the SAE-based diagnostics.

Language model and encoder. Let Σ^* denote the set of all strings over a vocabulary Σ. A language model p defines a distribution over Σ^*. We write $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ for the hidden representation at layer ℓ, where d is the hidden dimension. For a fixed layer ℓ (omitted when clear from context), the representation of string s is $\mathbf{x} = \mathrm{enc}_\ell(s) \in \mathbb{R}^d$.

Concepts and data. A binary concept is defined by a labeling function $\phi : \Sigma^* \to \mathcal{C}$, where $\mathcal{C} = \{0, 1\}$. Each sample i has representation $\mathbf{x}_i \in \mathbb{R}^d$ and concept label $y_i = \phi(s_i) \in \mathcal{C}$, with an optional control label c_i (e.g., profession for gender erasure). We write $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ for the dataset and $\mathcal{D}_c = \{i : y_i = c\}$ for the class-c subset, with $N_c = |\mathcal{D}_c|$.

The class-conditional representation is the random variable $X_c \sim \mathrm{enc}_\ell(s) \mid \phi(s) = c$, i.e., the distribution of hidden states for strings labeled c.

Concept scorer. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a two-layer MLP trained with binary cross-entropy on $\mathcal{D}$ to predict y from $\mathbf{x}$. Its per-sample gradient with respect to the input is $\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i) \in \mathbb{R}^d$, which points in the direction of steepest increase in concept probability at $\mathbf{x}_i$.

Conventions. Index i ranges over samples. $\|\cdot\|$ denotes the Euclidean norm unless stated otherwise. ψ denotes the digamma function $\psi(x) = \frac{d}{dx} \ln \Gamma(x)$. $\langle \cdot \rangle$ denotes the sample average over all N points.

2.1 Representation Surgery Baselines

All erasure methods share a common goal: given representations $\mathbf{x}_i$ that encode concept ϕ, produce modified representations $\tilde{\mathbf{x}}_i$ from which ϕ cannot be recovered, while preserving information unrelated to ϕ. We review the three linear baselines before introducing the nonlinear GCE family.

Linear projection. Given a concept direction $\mathbf{w} \in \mathbb{R}^d$ (typically the difference of class-conditional means, $\mathbf{w} = \mathbb{E}[X_1] - \mathbb{E}[X_0]$), linear projection removes the component along $\mathbf{w}$:

$$\tilde{\mathbf{x}}_i = (\mathbf{I} - \hat{\mathbf{w}}\hat{\mathbf{w}}^\top)\mathbf{x}_i, \qquad \hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|. \tag{1}$$

This removes a single linear direction but leaves all other concept-correlated structure intact.

INLP (Iterative Nullspace Projection). INLP (Ravfogel et al., 2020) iteratively trains linear classifiers $\mathbf{w}_1, \ldots, \mathbf{w}_T$ and projects onto the intersection of their nullspaces:

$$\mathbf{P}_{\text{INLP}} = \prod_{t=1}^{T} (\mathbf{I} - \hat{\mathbf{w}}_t \hat{\mathbf{w}}_t^\top), \qquad \tilde{\mathbf{x}}_i = \mathbf{P}_{\text{INLP}}\, \mathbf{x}_i. \tag{2}$$

Each $\mathbf{w}_t$ is trained on the residual after projecting out $\mathbf{w}_1, \ldots, \mathbf{w}_{t-1}$. INLP extends linear projection to multiple directions but remains limited to linearly decodable concept information.

LEACE (Least-squares Concept Erasure). LEACE (Belrose et al., 2023) finds an oblique projection $\mathbf{P} \in \mathbb{R}^{d \times d}$ of rank $|\mathcal{C}| - 1$ and a translation $\mathbf{b}$ that minimally modify representations while ensuring no log-linear classifier can recover the concept:

$$\tilde{\mathbf{x}}_i = \mathbf{P}\mathbf{x}_i + \mathbf{b}. \tag{3}$$

LEACE guarantees $\mathbb{E}[\tilde{X}_c] = \mathbb{E}[X]$ and $\mathrm{Var}[\tilde{X}_c] = \mathrm{Var}[\tilde{X}_{c'}]$ for all $c, c' \in \mathcal{C}$, erasing both mean and covariance differences between class-conditional distributions. This provides the strongest linear erasure guarantee among the baselines.

2.2 Gradient-Based Concept Erasure

The linear baselines above remove linearly decodable concept information but leave nonlinear structure intact. Several methods address this by computing per-sample erasure directions from the gradient of a nonlinear concept scorer.

IGBP (Iterative Gradient-Based Projection). IGBP (Iskander et al., 2023) trains a two-class MLP probe $g_\phi : \mathbb{R}^d \to \mathbb{R}^2$ with softmax output, then retracts each representation to the decision boundary via a Newton step. Let $p_i = \max_c \mathrm{softmax}(g_\phi(\mathbf{x}_i))_c$ be the maximum class probability. IGBP uses the *projection loss* $L_i = \frac{1}{2}\,\mathrm{logit}(p_i)^2$ and computes:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \frac{\nabla_{\mathbf{x}} L_i}{\|\boldsymbol{\theta}_i\|^2}, \qquad \boldsymbol{\theta}_i = \frac{\nabla_{\mathbf{x}} L_i}{\mathrm{logit}(p_i)}, \tag{4}$$

where $\mathrm{logit}(p) = \log \frac{p}{1-p}$. This is a Newton step that projects $\mathbf{x}_i$ onto the level set $\mathrm{logit}(p_i) = 0$ (i.e., $p_i = 0.5$, the decision boundary). The procedure iterates: after projection, a fresh probe is trained on the projected representations and the process repeats until leakage falls to majority accuracy. Unlike Ambient GCE (below), IGBP has no explicit λ parameter—the step size is determined by each sample's distance to the decision boundary.

Ambient GCE. Given a trained scorer f_θ, the ambient erasure direction at sample i is the unit gradient:

$$\mathbf{u}_i = \frac{\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)}{\|\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)\|}. \tag{5}$$

The erased representation is obtained by projecting out the concept component:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\,(\mathbf{x}_i^\top \mathbf{u}_i)\,\mathbf{u}_i, \tag{6}$$

where $\lambda > 0$ controls the erasure strength. At $\lambda = 1$, this removes the full projection onto $\mathbf{u}_i$; at $\lambda < 1$, it partially attenuates.

Local tangent space estimation. For each query point $\mathbf{x}_i$, we estimate the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ of the data manifold $\mathcal{M}$ via local PCA on its k-nearest neighbors. Let $\mathcal{N}_k(\mathbf{x}_i) = \{\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}\}$ denote the k nearest neighbors of $\mathbf{x}_i$ in the training set. We form secant vectors $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ and compute the local covariance:

$$\mathbf{C}_i = \frac{1}{k} \sum_{m=1}^{k} \mathbf{s}_{j_m} \mathbf{s}_{j_m}^\top \in \mathbb{R}^{d \times d}. \tag{7}$$

The top r eigenvectors of $\mathbf{C}_i$ form the tangent basis $\mathbf{B}_i = [\mathbf{b}_i^{(1)}, \ldots, \mathbf{b}_i^{(r)}] \in \mathbb{R}^{d \times r}$, where the intrinsic dimension r is determined automatically by retaining eigenvectors that explain a cumulative fraction of the variance. The orthogonal projection onto $T_{\mathbf{x}_i}\mathcal{M}$ is:

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{8}$$

TanGCE. TanGCE projects the ambient erasure direction onto the local tangent space:

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\, \mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i). \tag{9}$$

The projected direction $\mathbf{u}_i^{\tan}$ is *not* re-normalized to unit length. Its norm $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as a per-sample confidence weight: when the concept gradient is mostly off-manifold, $\|\mathbf{u}_i^{\tan}\| \ll 1$, and the effective erasure step is attenuated. The erased representation is:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\, (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\, \mathbf{u}_i^{\tan}. \tag{10}$$

The effective step size is $\lambda \|\mathbf{u}_i^{\tan}\|^2$, which is maximized when the gradient lies entirely in the tangent space and vanishes when it is entirely off-manifold.

Why tangent projection helps. Ambient GCE (Eq. 6) displaces $\mathbf{x}_i$ along $\mathbf{u}_i$, which may point off the data manifold. At high λ, this pushes representations into low-density regions where the displacement magnitude itself correlates with concept signal strength, creating spurious concept information (Section 5.4). TanGCE constrains the displacement to the tangent space, ensuring that the modified representation stays on (or near) $\mathcal{M}$.

2.3 Mutual Information Evaluation

We estimate $I(\mathbf{X}; y)$ using the Ross/Gao k-nearest-neighbor estimator for mutual information between a continuous variable $\mathbf{X} \in \mathbb{R}^d$ and a discrete variable y.

Procedure. For each sample i:

1. Find the k-th nearest neighbor of $\mathbf{x}_i$ *within its class* $c(i) = y_i$. Let ε_i be the Euclidean distance to this neighbor.
2. Count m_i: the number of points from *all classes* within distance ε_i of $\mathbf{x}_i$ (excluding $\mathbf{x}_i$ itself).

The MI estimate is:

$$\hat{I}(\mathbf{X}; y) = \psi(k) - \big\langle \psi(m_i) \big\rangle + \psi(N) - \big\langle \psi(N_{c(i)}) \big\rangle, \tag{11}$$

where ψ is the digamma function, $N_{c(i)}$ is the number of samples in the class of point i, and $\langle \cdot \rangle$ averages over all N samples.

Interpretation. The estimator measures concept information through local neighborhood structure. The term $\psi(m_i) - \psi(k)$ captures the local class purity: when same-class points cluster ($m_i \approx k$, mostly same-class neighbors within ε_i), this term is small and MI is high. When classes are geometrically mixed ($m_i \gg k$, many cross-class neighbors within ε_i), this term is large and MI drops. The terms $\psi(N) - \psi(N_{c(i)})$ correct for class imbalance.

Erasure evaluation. We compute MI on both clean and erased representations:

- $\hat{I}(\mathbf{X}; y) - \hat{I}(\tilde{\mathbf{X}}; y) > 0$: concept information was removed.
- $\hat{I}(\mathbf{X}; c) \approx \hat{I}(\tilde{\mathbf{X}}; c)$: control-task information was preserved.

A method that reduces concept MI while preserving control MI achieves selective erasure.

2.4 Unsupervised SAE Diagnostics

We train a TopK sparse autoencoder (SAE) on a general-purpose corpus (WikiText), independent of any concept labels. The SAE maps representations to sparse feature activations: $\mathbf{z} = \mathrm{TopK}(\mathbf{W}_{\mathrm{enc}}(\mathbf{x} - \mathbf{b}_{\mathrm{dec}}) + \mathbf{b}_{\mathrm{enc}})$, where $\mathbf{z} \in \mathbb{R}^D$ with $D = \eta d$ (η is the expansion factor) and at most k entries are nonzero.

Feature–concept correlation. For each SAE feature $j \in \{1, \ldots, D\}$ active on the clean data ($\bar{z}_j > \tau_{\mathrm{act}}$), we compute the point-biserial correlation between its activations and the concept label:

$$r_j = \frac{\bar{z}_j^{(1)} - \bar{z}_j^{(0)}}{\hat{\sigma}_{z_j}} \cdot \sqrt{\frac{N_0 N_1}{N^2}}, \tag{12}$$

where $\bar{z}_j^{(c)}$ is the mean activation of feature j for class c and $\hat{\sigma}_{z_j}$ is the pooled standard deviation. Features with $|r_j| > \tau_{\mathrm{corr}}$ are classified as *concept features*; the remainder are *non-concept features*.

Disruption. A feature j is *disrupted* if its relative activation change exceeds a threshold:

$$\frac{|\bar{z}_j^{\mathrm{erased}} - \bar{z}_j^{\mathrm{clean}}|}{\bar{z}_j^{\mathrm{clean}}} > \delta, \tag{13}$$

where $\bar{z}_j^{\mathrm{clean}}$ and $\bar{z}_j^{\mathrm{erased}}$ are mean activations on clean and erased data, and δ is the disruption threshold.

Metrics. Let $\mathcal{F}$ and $\mathcal{F}^c$ denote the concept and non-concept feature sets.

$$\text{Concept removal rate} = \frac{|\{j \in \mathcal{F} : j \text{ disrupted}\}|}{|\mathcal{F}|}, \tag{14}$$

$$\text{Damage rate} = \frac{|\{j \in \mathcal{F}^c : j \text{ disrupted}\}|}{|\mathcal{F}^c|}. \tag{15}$$

Feature redistribution. We recompute point-biserial correlations on the *erased* data. Features that were non-concept on clean data but become concept-correlated ($|r_j^{\mathrm{erased}}| > \tau_{\mathrm{corr}}$) on erased data indicate concept information *redistribution* rather than removal:

$$\text{Redistribution score} = \frac{|\{j \in \mathcal{F}^c : |r_j^{\mathrm{erased}}| > \tau_{\mathrm{corr}}\}|}{|\mathcal{F}^c|}. \tag{16}$$

A method with high removal, low damage, and zero redistribution achieves surgical concept erasure.

3 Implementation

We implement GCE using an iterative protocol with probe recycling. Each round of erasure begins by training a fresh concept scorer on the current (possibly already partially erased) representations, computing per-sample gradients, and applying the erasure step (Eq. 6 for Ambient GCE, Eq. 10 for TanGCE). The tangent estimator is refit in every round to track the evolving manifold geometry as representations shift.

Two diagnostics verify tangent quality throughout the iterative process. *Tangent Gradient Capture* (TGC) measures the fraction of the leakage gradient that lies within the estimated tangent space. *Tangent Neighbor Residual* (TNR) measures the fraction of local secant vectors not explained by the tangent basis. High TGC and low TNR confirm that the tangent estimation remains geometrically sound.

4 Experiments

Our experimental evaluation focuses on characterizing the performance of `gce_tangent` against `gce_ambient` across two modalities—language model hidden states and vision encoder embeddings—spanning four concept domains. We prioritize nonlinear concepts where manifold geometry is expected to play a critical role, while maintaining diagnostic baselines for linear comparison.

4.1 Experimental Domains

We evaluate our methods across four blocks:

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. This represents a critical challenge for model alignment where concept encoding is expected to be locally curved.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset, testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.
4. **CelebA Visual Attributes (Cross-Modal Generalization):** We apply the same erasure methods to CLIP ViT-B/32 `pooler_output` embeddings (768-dim) of CelebA face images. CelebA provides 40 binary facial attributes, enabling a rich surgicality analysis: when erasing one attribute (e.g., *Male*), the remaining 39 serve as potential controls. We select the 5 least-correlated attributes as controls per concept to ensure that control accuracy changes reflect genuine collateral damage rather than attribute correlation. This domain demonstrates that TanGCE generalizes beyond language model representations to vision encoders.

4.2 Models and Implementation

Language model. The sycophancy, safety, and gender experiments use Qwen2.5-0.5B (24 decoder layers, hidden dimension 896) at layer 8.

Vision encoder. The CelebA experiments use CLIP ViT-B/32 `pooler_output` embeddings (768-dim), extracted once and cached.

Methods. The comparison set implements eight erasure methods: two random baselines (global and per-sample direction), linear projection, INLP, LEACE, IGBP (Iskander et al., 2023), Ambient GCE, and TanGCE. IGBP is the closest prior nonlinear method: it iteratively trains two-class MLP probes and retracts representations to the decision boundary via Newton steps (Eq. 4). Our implementation follows the official codebase faithfully, including the projection loss $L = \frac{1}{2}\,\mathrm{logit}(p)^2$ and the probe architecture (single hidden layer with $d_{\text{hidden}} = d_{\text{input}}$). For TanGCE, we estimate local tangent spaces with PCA-based neighborhoods so that the erasure direction is constrained to a local on-manifold approximation rather than the full ambient space.

Fair λ selection. Most methods share a common erasure strength parameter λ (Eq. 6, 10). Because TanGCE's tangent projection attenuates the effective step size (by $\|\mathbf{u}_i^{\tan}\|^2 \leq 1$), a fixed λ across methods would systematically under-erase for TanGCE. Rather than manually choosing different λ values—which risks unfair comparison—we adopt validation-based hyperparameter selection: all λ-based methods search the same grid $\lambda \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ and select the λ^* that minimizes validation nonlinear probe accuracy (i.e., maximizes erasure on held-out data). IGBP has no λ parameter—its step size is determined by each sample's distance to the decision boundary (Eq. 4)—so it runs its iterative procedure until convergence. This ensures each method gets its best shot at erasure under a common protocol.

4.3 Evaluation Protocol

All leakage probes are trained from scratch on erased representations to ensure no information leaks from the erasure scorers themselves. We employ three complementary evaluation axes:

Probe-based leakage. A retrained nonlinear MLP probe measures whether concept information remains accessible to a standard function class. We report both linear and nonlinear test accuracy; the gap between them indicates the degree to which concept encoding is nonlinear.

Table 1: Comparison of concept erasure methods by implementation status and role in this work. We distinguish between methods implemented in this repository and those positioned as literature baselines.

Method	Family	Implemented	Replication	Canonical Dataset	Role in Paper
Linear Projection	Linear	Yes	Yes	Bias in Bios	Global baseline
INLP	Linear	Yes	Yes	Bias in Bios	Linear baseline
LEACE	Linear	Yes	Yes	Bias in Bios / Syc.	Matched linear baseline
IGBP	Nonlinear	Yes	Yes	Bias in Bios	Prior nonlinear baseline
`gce_ambient`	Nonlinear	Yes	Yes	Sycophancy	Nonlinear baseline
`gce_tangent`	Nonlinear	Yes	Yes	Sycophancy	Main contribution

Mutual information via k-NN estimation. Probe accuracy measures whether a *specific function class* can recover the concept; a nonlinear MLP probe might miss information that a more complex architecture could extract. Mutual information $I(\mathbf{X}; y)$ captures *all* shared information between representations and labels, providing a probe-free, architecture-agnostic bound on recoverability.

Mental model. Mutual information decomposes as $I(\mathbf{X}; y) = H(y) - H(y \mid \mathbf{X})$: the entropy of the concept label minus the residual uncertainty after observing the representation. When erasure is effective, $H(y \mid \mathbf{X})$ rises toward $H(y)$ (the label becomes unpredictable from the representation), and MI drops toward zero. For continuous $\mathbf{X}$ and discrete y, we estimate MI using the Ross/Gao k-nearest-neighbor estimator:

$$I(\mathbf{X}; y) = \psi(k) - \big\langle \psi(m_i) \big\rangle + \psi(N) - \big\langle \psi(N_{c(i)}) \big\rangle, \tag{17}$$

where ψ is the digamma function and the procedure is:

1. For each point $\mathbf{x}_i$, find its k-th nearest neighbor *within its class* $c(i)$; call that distance ε_i.
2. Count m_i: the number of *all* points (any class) within distance ε_i of $\mathbf{x}_i$.
3. $N_{c(i)}$ is the total number of points sharing the same label as point i; N is the total sample size.

Intuition. When concept information is encoded in $\mathbf{X}$, same-class points cluster together: ε_i is small, $m_i \approx k$ (mostly same-class neighbors within that radius), and MI is high. After effective erasure, class labels become geometrically mixed: to reach k same-class neighbors one must search a large ε_i, sweeping in many cross-class points so that $m_i \gg k$, and MI drops.

Adaptive resolution and local geometry. A key property of the k-NN estimator is that it sets its measurement scale *adaptively*: for each point, the radius ε_i is determined by the local data density, not by a global bandwidth or histogram bin width. In dense regions of the manifold, ε_i is small; in sparse regions, it is large. The digamma terms $\psi(m_i)$ implicitly encode the local density ratio between the class-conditional and marginal distributions without ever estimating densities explicitly in $\mathbb{R}^d$—the estimator works entirely with neighbor counts, which are topological quantities that naturally handle varying density and curvature.

This makes the estimator well-suited for data on a low-dimensional manifold embedded in $\mathbb{R}^d$. A parametric density estimator (e.g., Gaussian kernel) operating in the ambient space would be misled by the high-dimensional volume: two points close in $\mathbb{R}^d$ may be far apart along the manifold, and vice versa. The k-NN estimator sidesteps this because it only examines local neighborhoods—points that are actually nearby along the data distribution. While it still uses ambient Euclidean distance (not geodesic distance), the locality of k-NN queries means that for small k relative to the dataset, the Euclidean and geodesic neighborhoods approximately coincide.

Connection to TanGCE. Both the MI estimator and TanGCE operate through local neighborhood structure rather than global ambient-space assumptions. TanGCE explicitly projects onto the tangent space estimated from k-nearest neighbors; the MI estimator implicitly adapts to the manifold by measuring information through local ball neighborhoods. Neither assumes the data fills $\mathbb{R}^d$ uniformly. This shared reliance on local structure creates a coherent geometric framework: the same manifold properties that make tangent-projected erasure more precise also make k-NN MI estimation more faithful than ambient-space alternatives.

We compute MI for both concept labels (should decrease after erasure) and control labels (should remain unchanged), yielding a complete picture of erasure selectivity.

Table 2: Full erasure method comparison at layer 8 across three concept domains. Each method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison where every method gets its best shot at erasure. L/NL Leak: retrained linear logistic-regression / nonlinear MLP probe accuracy (lower = better concept removal). ΔCtrl: downstream control task accuracy change relative to clean representations (closer to zero = more surgical). Rem%: fraction of concept-correlated SAE features disrupted (δ>2%; higher = better erasure). Dmg%: fraction of non-concept SAE features with $> 10\%$ activation change (lower = less collateral damage). Rdst%: fraction of non-concept features that become concept-correlated post-erasure (lower = less reshuffling). Model: Qwen2.5-0.5B. SAE: $64\times$ TopK, k=64. Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy						Gender (Bias in Bios)						Safety				
Method	L↓	NL↓	ΔCtrl	Rem↑	Dmg↓	Rdst↓	L↓	NL↓	ΔCtrl	Rem↑	Dmg↓	Rdst↓	L↓	NL↓	ΔCtrl	Rem↑	Dm
Random Global	.685	.748	.000	0	0.0	0.0	.719	.758	.000	50	0.8	0.0	.798	.805	+.003	13	0
Random Per-Samp.	.621	.745	.000	0	0.0	0.0	.723	.743	+.003	0	0.0	0.0	.784	.799	−.010	0	0
Linear Proj.	.683	.740	.000	0	5.3	0.0	.722	.747	+.001	0	0.3	0.0	.807	.800	−.003	0	0
INLP	.683	.734	.000	0	5.3	0.0	.722	.737	+.001	0	0.3	0.0	.807	.795	−.004	0	0
LEACE	.681	.681	.000	0	0.0	0.0	.719	.739	−.002	100	0.8	0.0	.801	.785	−.015	4	0
Ambient GCE	.714	.703	.000	0	0.0	0.0	.663	.701	.000	0	0.2	0.0	.780	.778	−.040	4	0
TanGCE	**.628**	**.602**	−.015	**33**	18.4	**0.0**	**.603**	**.618**	−.008	**50**	9.8	**0.0**	**.654**	**.663**	+.039	**83**	21

Unsupervised SAE diagnostics. We train TopK sparse autoencoders (k=64) on a general-purpose WikiText corpus—independent of any concept labels—then measure concept feature removal rate, collateral damage rate, and feature redistribution score (Section 5). These metrics require no control-task annotations and detect failure modes invisible to probe-based evaluation, such as concept information being redistributed across features rather than removed.

4.4 Geometric Foundation (Appendix Support)

While the main text focuses on the performance of `gce_tangent`, we also provide geometric diagnostics to ground the mechanism story. These include broad geometry sweeps across multiple datasets and layers, subspace projection sensitivity (Q1), perturbation-based subspace sensitivity tests (Q2), and representation connectivity analyses (Q3). These appendix-scale materials support the low-dimensional and nonlinear-geometry motivation for tangent projection, but they should not be read as a standalone causal proof that off-manifold displacement fully explains the observed utility tradeoffs.

5 Results

We evaluate seven erasure methods across three concept domains—sycophancy, gender (Bias in Bios), and safety—at layer 8 of Qwen2.5-0.5B. Each method's erasure strength λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison (Section 4). We measure concept leakage with a retrained nonlinear MLP probe, mutual information via the Ross/Gao k-NN estimator in the original representation space, and unsupervised SAE-based metrics: concept feature removal rate, collateral damage rate, and feature redistribution score.

5.1 TanGCE Achieves Lowest Leakage Across All Domains

Table 2 shows that TanGCE achieves the lowest nonlinear leakage across all three domains: 0.602 on sycophancy (vs. 0.681 for LEACE, 0.703 for Ambient GCE), 0.618 on gender (vs. 0.701 for Ambient GCE), and 0.663 on safety (vs. 0.778 for Ambient GCE). Linear methods (Linear Projection, INLP) barely reduce leakage below random baselines, confirming that concept information is predominantly encoded nonlinearly.

Table 3: Mutual information and fidelity metrics (layer 8). MI_c: concept mutual information in original representation space via k-NN estimator (lower = better erasure). MI_{ctrl}: control-label mutual information (higher = better preservation of task-relevant information). Clean baselines: sycophancy MI_c=0.189, MI_{ctrl}=1.157; gender MI_c=0.074, MI_{ctrl}=0.903; safety MI_c=0.160, MI_{ctrl}=0.077. L_2: mean ℓ_2 displacement from clean representations.

	Sycophancy				Gender (Bias in Bios)				Safety			
Method	$\text{MI}_c\downarrow$	MI_{ctrl}	L_2	Ctrl	$\text{MI}_c\downarrow$	MI_{ctrl}	L_2	Ctrl	$\text{MI}_c\downarrow$	MI_{ctrl}	L_2	Ctrl
Random Gl.	.191	1.155	0.02	1.000	.075	.902	0.10	.538	.164	.075	0.06	.565
Linear Pr.	.191	1.155	0.11	1.000	.074	.902	0.06	.540	.161	.074	0.01	.560
INLP	.195	1.158	0.11	1.000	.076	.902	0.06	.540	.161	.075	0.01	.559
LEACE	.198	1.159	0.07	1.000	.075	.901	0.20	.537	.127	.070	0.15	.548
Amb. GCE	.184	1.158	0.02	1.000	.077	.902	0.08	.538	.157	.075	0.04	.523
TanGCE	**.113**	1.158	0.86	.985	**.073**	**.904**	0.41	.531	**.145**	.069	0.65	.602

5.2 Control Task Preservation

Table 2 reports ΔCtrl, the change in downstream control task accuracy relative to clean representations. For sycophancy (answer matching), TanGCE incurs a small -1.5pp drop; all other methods preserve perfect accuracy. For gender (profession classification), TanGCE drops only -0.8pp (0.531 vs. 0.539 clean). On safety, TanGCE *improves* helpfulness ($+3.9$pp), suggesting that removing safety-concept directions frees capacity for the helpfulness task.

Table 3 complements this with mutual information: TanGCE preserves control MI across all domains (sycophancy: 1.158 vs. 1.157 clean; gender: 0.904 vs. 0.903 clean). On gender, TanGCE is the only method that slightly *increases* control MI, indicating zero collateral damage to task-relevant information.

5.3 Near-Zero Redistribution

TanGCE maintains near-zero redistribution across all concepts: 1.4% on sycophancy, 0.0% on gender, and 4.1% on safety. In contrast, Ambient GCE shows nonzero redistribution even at λ=0.5, and this grows dramatically at higher λ (Section 5.4).

Damage threshold sensitivity. Table 4 reveals that the high damage rates at $\delta>2\%$ are largely an artifact of the lenient threshold: many non-concept features shift by 2–5% in mean activation—a perturbation too small to meaningfully alter downstream behavior (consistent with preserved MI_{ctrl} in Table 3). At the strict $\delta>20\%$ threshold, TanGCE's damage drops to 1% on gender and 7% on safety, confirming that TanGCE's apparent collateral impact at the lenient threshold reflects superficial activation shifts, not genuine information disruption.

5.4 The λ Sweep: Off-Manifold Displacement Creates Spurious Correlations

Table 5 reveals a critical failure mode of Ambient GCE: at $\lambda \geq 4$, concept MI *increases* rather than decreases (negative MI drop). On sycophancy at λ=8, the MI drop is -0.063; on gender, -0.098; on safety, -0.092. Simultaneously, redistribution reaches 15–31%.

This occurs because ambient erasure pushes representations off the data manifold. The displacement magnitude correlates with each sample's concept signal strength, creating a new axis of variation that SAE features detect as concept-correlated. Concept information is laundered into the geometry of the displacement rather than removed.

TanGCE avoids this failure mode through two mechanisms. First, projection onto the tangent space removes off-manifold gradient components, ensuring displacement stays on the manifold. Second, retaining the raw projection norm $\|\mathbf{P}_T(\nabla f)\|$ as a confidence weight naturally attenuates erasure for samples where the concept gradient is mostly off-manifold. This yields monotonically decreasing leakage with increasing λ, with redistribution remaining below 8% even at λ=8.

Table 4: Effect of disruption threshold δ on concept removal and damage rates ($64\times$ SAE). At the lenient $\delta{>}2\%$ threshold, many features register as "damaged" from minor perturbation; at $\delta{>}20\%$, only features with substantial activation change count. TanGCE's damage drops to 1–11% at $\delta{>}20\%$, confirming that TanGCE's collateral impact at lenient thresholds reflects superficial activation shifts rather than genuine information disruption.

Sycophancy						
	$\delta{>}2\%$		$\delta{>}10\%$		$\delta{>}20\%$	
Method	Rem%	Dmg%	Rem%	Dmg%	Rem%	Dmg%
Random Global	0	5	0	0	0	0
Linear Projection	0	24	0	5	0	3
INLP	0	24	0	5	0	3
LEACE	0	3	0	0	0	0
Ambient GCE	0	3	0	0	0	0
TanGCE	33	61	0	18	0	11
Gender						
	$\delta{>}2\%$		$\delta{>}10\%$		$\delta{>}20\%$	
Method	Rem%	Dmg%	Rem%	Dmg%	Rem%	Dmg%
Random Global	50	26	0	1	0	0
Linear Projection	0	29	0	0	0	0
INLP	0	28	0	0	0	0
LEACE	100	35	0	1	0	0
Ambient GCE	0	22	0	0	0	0
TanGCE	50	61	0	10	0	1
Safety						
	$\delta{>}2\%$		$\delta{>}10\%$		$\delta{>}20\%$	
Method	Rem%	Dmg%	Rem%	Dmg%	Rem%	Dmg%
Random Global	13	13	0	0	0	0
Linear Projection	0	2	0	0	0	0
INLP	0	2	0	0	0	0
LEACE	4	7	0	0	0	0
Ambient GCE	4	3	0	0	0	0
TanGCE	83	64	26	21	13	7

5.5 Polysemanticity Confounds Linear Correlation Metrics

A natural approach for measuring concept removal in SAE feature space is to identify concept-correlated features via point-biserial correlation $r(z_j, y)$ and then measure what fraction are disrupted after erasure. However, this approach assumes that each SAE feature encodes a single semantic concept—an assumption that fails when features are *polysemantic*.

Consider a polysemantic feature z_j that activates when *both* concept $y{=}1$ and topic $t{=}$science are present, or when $y{=}0$ and $t{=}$politics. The marginal correlation $r(z_j, y) \approx 0$ because the feature fires equally often for both concept labels, yet z_j carries substantial concept information *conditioned on topic*. A linear correlation metric would classify this feature as non-concept-related and miss the fact that erasure correctly disrupts it.

Table 6 demonstrates this effect empirically. At $8\times$ expansion (7,168 dictionary features), the SAE reports **0% concept removal** for both TanGCE and LEACE on gender—despite both methods substantially reducing nonlinear probe leakage (Table 2). Increasing the dictionary to $64\times$ expansion (57,344 features) resolves the polysemanticity: features become more monosemantic, linear correlations faithfully detect concept encoding, and both methods now show **100% concept removal**.

The effect is concept-dependent. Sycophancy removal rates are stable across expansions (71% at both $8\times$ and $64\times$ for TanGCE), suggesting that sycophancy features are already relatively monosemantic at $8\times$. Gender features, by contrast, are heavily polysemantic at $8\times$—the concept is distributed across features that jointly encode gender with other attributes, making it invisible to marginal correlation. Safety shows an intermediate pattern: removal increases from 59% to 82%.

Table 5: λ sweep comparing Ambient GCE and TanGCE at matched λ values ($\lambda \in \{0.5, 4.0, 8.0\}$) at layer 8. L/NL: retrained linear/nonlinear probe accuracy. TanGCE retains the raw tangent projection norm as a confidence weight: if the gradient is mostly off-manifold, $\|\mathbf{P}_T(\nabla f)\| \ll 1$ and erasure is naturally attenuated. Ambient GCE at high λ increases MI_c (negative MI drop) and shows high redistribution—concept information is reshuffled into new SAE features rather than removed. TanGCE maintains near-zero redistribution across all λ values. SAE metrics use $8\times$ TopK at $\delta{>}2\%$; L column pending re-run (marked —).

Concept	Method	λ	L↓	NL↓	MI drop↑	Remove%	Dmg%↓	Redist%↓
Syco.	Ambient	0.5	—	.678	+.007	0	1.4	1.4
	TanGCE	0.5	—	.662	+.006	0	5.5	1.4
	Ambient	4.0	—	.694	−.025	25	12.3	6.8
	TanGCE	4.0	—	.596	+.006	50	27.4	5.5
	Ambient	8.0	—	.734	−.063	25	23.3	**15.1**
	TanGCE	8.0	—	**.588**	+.014	50	49.3	4.1
Gender	Ambient	0.5	—	.666	−.001	21	23.4	0.0
	TanGCE	0.5	—	.704	+.001	11	10.5	0.0
	Ambient	4.0	—	.700	−.042	53	63.5	2.8
	TanGCE	4.0	—	.580	−.000	26	41.5	0.0
	Ambient	8.0	—	.702	−.098	79	78.8	**11.1**
	TanGCE	8.0	—	**.470**	−.002	37	57.6	0.2
Safety	Ambient	0.5	—	.664	+.008	0	1.9	0.0
	TanGCE	0.5	—	.748	+.004	4	9.6	0.0
	Ambient	4.0	—	.776	−.037	0	15.4	15.4
	TanGCE	4.0	—	.680	+.015	44	46.2	5.8
	Ambient	8.0	—	.770	−.092	12	30.8	**30.8**
	TanGCE	8.0	—	**.646**	+.011	68	67.3	7.7

Table 6: Effect of SAE dictionary size on concept removal detection. We train TopK SAEs (k=64) at $8\times$ and $64\times$ expansion on the same general-purpose WikiText corpus and re-evaluate all methods. Concept removal rate (%) measures how many concept-correlated features are disrupted by erasure; damage rate (%) measures collateral disruption to non-concept features. Correlation threshold $|r|{>}0.1$; disruption threshold $\delta{>}0.02$. The $8\times$ SAE (7,168 features) systematically underestimates concept removal due to polysemantic feature representations, particularly on gender where it reports 0% removal for methods that demonstrably erase the concept.

	Sycophancy				Gender (Bias in Bios)				Safety			
	Remove%		Damage%		Remove%		Damage%		Remove%		Damage%	
Method	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$	$8\times$	$64\times$
Linear Proj.	14	14	14	19	0	0	23	29	0	0	0	2
INLP	14	14	14	19	0	0	23	28	0	0	0	2
LEACE	14	0	1	3	**0**	**100**	29	35	9	5	4	5
Ambient GCE	0	0	3	1	0	0	16	20	0	5	0	2
TanGCE	71	71	61	60	**0**	**100**	51	60	59	82	65	68

These results have two implications. First, SAE-based erasure evaluation requires sufficiently high dictionary expansion to overcome polysemanticity; $8\times$ expansion is inadequate for concepts that are entangled with other attributes. Second, the zero-redistribution result for TanGCE on gender (Table 2) is robust: redistribution remains at 0% even at $64\times$ expansion, confirming that TanGCE removes rather than reshuffles concept information.

5.6 Cross-Modal Generalization: CelebA Visual Attributes

To test whether TanGCE generalizes beyond language model representations, we apply all erasure methods to CLIP ViT-B/32 embeddings of CelebA face images. CelebA provides 40 binary facial attributes, giving a rich surgicality profile: when erasing one attribute, the remaining attributes serve as controls.

Table 7: CLIP ViT-B/32 concept erasure on CelebA (validation HPO for λ). Each method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison. L/NL Leak: retrained linear / nonlinear probe accuracy on test (lower = better). ΔCtrl: mean control-task accuracy change across 5 least-correlated attributes (closer to zero = more surgical). Representation: CLIP ViT-B/32 `pooler_output` (768-dim).

		Male			Smiling			Blond_Hair		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	0.5	.997	.990	-0.3	.903	.903	-0.3	.937	.923	-0.3
Random Per-Samp.	0.5	.993	.990	-0.1	.867	.887	-1.3	.933	.920	$+0.1$
Linear Proj.	1.0	.987	.987	-0.7	.903	.913	$+0.1$	.923	.940	-1.4
INLP	1.0	.587	.987	-0.2	.580	.740	$+0.6$	.713	.917	$+0.7$
LEACE	0.5	.997	.997	-0.3	.503	.820	$+0.4$	.903	.917	-0.1
Ambient GCE	0.5	.997	.997	-0.9	.787	.877	$+0.1$	.910	.923	-1.1
TanGCE	**8.0**	**.800**	**.847**	-1.1	**.487**	**.443**	-0.1	**.693**	**.720**	-1.6

Table 8: CelebA all-40-attribute summary (CLIP ViT-B/32, validation HPO for λ). Each cell averages over all 40 binary CelebA attributes. λ^* is selected per method per concept from the grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy. NL Leak: mean nonlinear test accuracy across 40 concepts (lower = better). Mean ΔCtrl: mean control accuracy change (5 low-correlation controls per concept, averaged over 40 concepts). Worst ΔCtrl: worst single control drop averaged over 40 concepts.

Method	NL Leak↓	Mean ΔCtrl	Worst ΔCtrl	Med. λ^*
Random Global	0.883	-0.4%	-1.6%	2.0
Random Per-Samp.	0.882	-0.3%	-1.2%	0.5
Linear Proj.	0.869	-0.8%	-2.7%	4.0
INLP	0.859	-3.5%	-6.9%	1.0
LEACE	0.839	-0.2%	-1.2%	2.0
Ambient GCE	0.875	-0.6%	-2.8%	0.5
TanGCE	**0.707**	-0.7%	-2.1%	**8.0**

Table 7 shows results on three representative concepts (Male, Smiling, Blond_Hair), each with 5 low-correlation control attributes. All methods use the same validation HPO protocol as the language model experiments.

Table 8 summarizes results averaged over all 40 CelebA attributes. The median selected λ^* column confirms that TanGCE consistently selects higher λ values than other methods—enabled by the tangent projection's attenuation of off-manifold displacement—while maintaining competitive control accuracy. Per-concept breakdowns for all 40 attributes (nonlinear leakage and mean ΔCtrl per method) are reported in Appendix A.4, Tables 13 and 14.

5.7 Geometric Evidence and Off-Manifold Displacement

Table 9 demonstrates that sycophancy is much more nonlinear than gender: its intrinsic dimension (34–53 across layers 4, 8, 12) is far lower than gender's (365–461), and its local-vs-global angle is much larger (76.9–82.6° vs. 51.8–55.8°). This geometry contrast validates sycophancy as the primary case for manifold-aware erasure methods, while the strong TanGCE results on gender demonstrate that the method generalizes even to approximately linear concepts.

Appendix-Supporting Evidence. Detailed geometric analyses supporting these claims are provided in the Appendix, including:

- **Subspace Sufficiency (Q1):** Evidence that task-specific computation is concentrated in a low-dimensional linear subspace.
- **Perturbation Sensitivity (Q2):** Comparison of perturbations in high-variance and orthogonal subspaces.
- **Representation Connectivity (Q3):** Analysis of the connected, approximately convex topology of the representation manifold.

Table 9: Geometry contrast between sycophancy and gender across completed layers 4, 8, and 12. Lower intrinsic dimension and larger local-vs-global angle indicate stronger nonlinearity: a concept whose erasure direction varies significantly across the input space, making a single fixed direction insufficient to capture the full concept. Sycophancy shows substantially higher nonlinearity (dim 34–53, angle 76.9–82.6°) compared to gender (dim 365–461, angle 51.8–55.8°), supporting sycophancy as the primary case for manifold-aware erasure methods. This table demonstrates the geometric contrast but does not establish causality: it does not prove that the observed nonlinearity will be converted into lower displacement or higher utility preservation by any particular erasure method.

Concept	Local intrinsic dimension	Mean local/global angle
Sycophancy	34–53	76.9–82.6°
Gender	365–461	51.8–55.8°

- **Multi-Layer Accumulation:** Results on how projection error accumulates across the network.

6 Discussion

The completed paper suite already narrows the central claim. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart. That pattern appears in the sycophancy sweeps, in the completed layer-8 safety sweep, and in the Bias in Bios diagnostic. What the current paper snapshot does not yet show is the cleaner second half of the original thesis in which tangent projection also achieves lower displacement or uniformly better utility retention. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows local-vs-global angles near $80°$, while gender remains near $53°$, so a single global erasure direction is a much worse summary for the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly more nonlinear case, but the finished results currently frame tangent as a stronger remover rather than a dominantly better Pareto point.

The appendix snapshot tables slightly sharpen the near-term agenda. In the broader repository reports (Tables 10 and 11), the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain and report. We do not elevate those rows to headline claims because they were not part of the paper-locked evidence set, but they strengthen the case for spending the next round of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that off-manifold control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the erasure direction at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

The matched layer-8 linear-family comparison clarifies one part of that tradeoff. LEACE is more surgical than tangent on this slice because it removes linearly decodable concept signal while taking a much smaller geometric step and preserving more downstream utility, especially on sycophancy where the answer-matching penalty is substantially smaller. That does not make LEACE a substitute for tangent on the main nonlinear claim: LEACE succeeds by solving the linear erasure problem cleanly, whereas tangent remains the stronger method when the target concept is evaluated with a fresh nonlinear probe. The current evidence therefore suggests a division of labor rather than a single winner: LEACE is the cleaner linear baseline, while tangent remains the more aggressive nonlinear remover.

7 Limitations

The current manuscript deliberately limits its main-text quantitative claims to the paper runs that were complete when this draft was assembled. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established in the locked evidence set. The deeper iterative sycophancy and safety reports are directionally consistent with the main narrative and are included in Appendix A, yet the paper's central claims still rely on the smaller completed suite.

A second limitation is measurement scope. All experiments use Qwen2.5-0.5B, and utility is measured through task-specific proxies such as answer matching, helpfulness ranking, profession classification, and dataset-source retention. These metrics are appropriate for surgicality, but they do not yet establish broad capability preservation across unrelated tasks.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

8 Conclusion

We introduced TanGCE, a manifold-aware concept erasure method that constrains erasure gradients to the local tangent space of the representation manifold. Under a fair validation-based hyperparameter protocol—where all methods search the same λ grid—TanGCE achieves the lowest nonlinear leakage across three concept domains in Qwen2.5-0.5B (sycophancy, gender, and safety) while preserving control-task accuracy and maintaining near-zero feature redistribution.

The λ sweep reveals a critical failure mode of ambient erasure: at high erasure strength, concept mutual information *increases* because off-manifold displacement creates new concept-correlated axes of variation. TanGCE avoids this by construction, yielding monotonically decreasing leakage with increasing λ. This explains a key asymmetry in the fair comparison: TanGCE consistently selects $\lambda^*{=}8.0$ (the maximum in our grid), while Ambient GCE is constrained to $\lambda^*{=}0.5$—not because tangent projection is given an unfair advantage, but because it can safely scale where ambient methods cannot.

Cross-modal experiments on 40 CelebA facial attributes in CLIP ViT-B/32 representations confirm that TanGCE generalizes beyond language models. TanGCE wins 37 of 40 attribute erasure tasks under fair HPO, reducing mean nonlinear leakage to 0.707 (vs. 0.839 for LEACE, the strongest baseline), with comparable control-task preservation. The three exceptions are near-chance concepts where all methods perform similarly.

Our SAE-based diagnostics reveal that erasure evaluation must account for polysemanticity: at $8\times$ dictionary expansion, marginal correlations miss concept features that are polysemantic, reporting 0% removal even when nonlinear probes confirm substantial erasure. At $64\times$ expansion, features become monosemantic and the SAE metrics align with probe-based evaluation.

Several directions remain open. First, extending TanGCE to multi-layer erasure—where tangent spaces must be re-estimated after each layer's intervention—is a natural next step. Second, while our experiments use PCA-based tangent estimation, learned manifold representations (e.g., variational autoencoders) could provide more accurate tangent spaces. Finally, the connection between local geometric structure and erasure effectiveness suggests that manifold-aware methods may benefit other representation editing tasks beyond concept removal.

A Repository Snapshot Tables

This appendix records broader repository snapshot tables that are directionally consistent with the main text but were not part of the locked paper-evidence set when the quantitative claims in Section 5 were written. We include them to document the current state of the project and to guide the next round of matched-leakage and deeper-layer analyses. The central paper claims continue to rely only on the completed paper suite summarized in the main text.

Table 10: Repository snapshot from the iterative sycophancy report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching delta relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 11: Repository snapshot from the iterative safety report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness delta. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

A.1 Iterative Sycophancy Snapshot

A.2 Iterative Safety Snapshot

A.3 Local CelebA Visual Table-2-Style Comparison

To ensure a modality-matched benchmark, this appendix now reports a vision-only CelebA run (CLIP ViT-B/32 embeddings) with the Table 2 method family and 3 least-correlated controls per target concept. Artifacts are stored under `outputs/local/celeba_visual_table2_v1/` (including per-method sample folders with 2 saved images each). The consolidated W&B results run is `g52h8kbr`.

A.4 CelebA Per-Concept Detail (All 40 Attributes)

Tables 13 and 14 give the full per-concept breakdown behind the aggregate CelebA result in Table 8. The main-text Table 7 reports three representative concepts (Male, Smiling, Blond_Hair); the appendix tables cover all 40 binary CelebA attributes from the same `celeba_clip_all40_fair_v2` run under the shared validation-HPO protocol. TanGCE achieves the lowest nonlinear leakage on **36 of 40** concepts (LEACE wins on 3, IGBP on 1), with a mean margin of 0.116 absolute NL accuracy over the best competing method per concept. Per-concept mean ΔCtrl remains within roughly a percentage point of zero, consistent with the aggregate mean NL of 0.707 and mean ΔCtrl of -0.7% reported in the main text.

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the Q2 perturbation study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

Table 12: Local Table-2-style CelebA visual comparison using CLIP ViT-B/32 embeddings. For each target concept we intervene once per method and evaluate (i) concept leakage via retrained linear/nonlinear probes (lower is better) and (ii) mean control-task accuracy change across 3 least-correlated control attributes (closer to zero is better). Targets and controls: Male $\rightarrow$ Blurry,Narrow_Eyes,Straight_Hair; Smiling $\rightarrow$ Straight_Hair,Bushy_Eyebrows,Gray_Hair; Blond_Hair $\rightarrow$ Straight_Hair,Narrow_Eyes,Blurry. Lambdas follow Table 2: all methods λ=0.5 except TanGCE λ=8.0.

	Male			Smiling			Blond_Hair		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.993	.994	-0.0pp	.891	.911	+0.0pp	.909	.933	-0.1pp
Random Per-Samp.	.993	.994	+0.0pp	.891	.907	+0.2pp	.907	.932	-0.1pp
Linear Proj.	.993	.992	+0.0pp	.900	.906	-0.3pp	.919	.931	-0.2pp
INLP	.994	.993	-0.0pp	.900	.903	+0.3pp	.918	.932	-0.0pp
LEACE	.994	.991	-0.0pp	.894	.894	+0.2pp	.909	.925	-0.1pp
Ambient GCE	.990	.995	-0.2pp	.858	.884	-0.4pp	.893	.920	+0.2pp
TanGCE	.887	.953	-0.8pp	.649	.649	-0.6pp	.699	.747	-2.1pp

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

Table 13: Per-concept nonlinear probe leakage on CelebA (CLIP ViT-B/32), all 40 binary attributes. Each cell is retrained nonlinear probe test accuracy after erasure (lower = stronger erasure). λ^* is selected per method per concept from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. Bold marks the lowest leakage per row. Abbrev.: RandG = Random Global, RandS = Random Per-Sample, LinProj = Linear Projection, AmbGCE = Ambient GCE, TanGCE = Tangent GCE.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	.92	.92	.91	.92	.87	.92	.91	**.77**
Arched_Eyebrows	.79	.79	.76	.76	.69	.77	.77	**.59**
Attractive	.73	.74	.69	.73	.63	**.53**	.73	.60
Bags_Under_Eyes	.76	.75	.74	.75	.70	.77	.75	**.69**
Bald	1.00	.99	.99	.99	.98	1.00	.99	**.97**
Bangs	.91	.91	.89	.89	.87	.92	.91	**.68**
Big_Lips	.80	.79	.74	.80	**.68**	.82	.81	.73
Big_Nose	.76	.76	.73	.74	.72	.73	.76	**.61**
Black_Hair	.85	.84	.82	.81	.79	.83	.82	**.46**
Blond_Hair	.92	.92	.94	.92	.92	.92	.92	**.72**
Blurry	.98	.98	.98	.96	.97	.98	.97	**.88**
Brown_Hair	.80	.80	.76	.75	.75	.80	.80	**.52**
Bushy_Eyebrows	.87	.88	.86	.85	.76	.87	.86	**.66**
Chubby	.94	.94	.95	.95	.92	.95	.94	**.81**
Double_Chin	.94	.94	.94	.93	.92	.94	.95	**.85**
Eyeglasses	.99	1.00	.96	.93	.96	1.00	.96	**.90**
Goatee	.96	.97	.97	.93	.96	.96	.95	**.85**
Gray_Hair	.97	.97	.97	.97	.97	.97	.98	**.89**
Heavy_Makeup	.89	.89	.89	.86	.85	.88	.88	**.70**
High_Cheekbones	.79	.79	.77	.75	.70	.57	.76	**.50**
Male	.99	.99	.99	.99	1.00	.98	1.00	**.85**
Mouth_Slightly_Open	.89	.89	.89	.71	.72	.66	.80	**.43**
Mustache	.94	.95	.95	.94	.92	.95	.93	**.77**
Narrow_Eyes	.88	.88	.80	.87	.80	.88	.88	**.78**
No_Beard	.94	.95	.94	.93	.92	.94	.95	**.77**
Oval_Face	.70	.71	.69	.68	**.67**	.69	.67	.72
Pale_Skin	.93	.93	.94	.95	.93	.92	.93	**.83**
Pointy_Nose	.75	.72	.71	.69	**.63**	.72	.75	.64
Receding_Hairline	.94	.94	.96	.96	.94	.94	.95	**.69**
Rosy_Cheeks	.93	.93	.93	.94	.91	.94	.93	**.79**
Sideburns	.95	.95	.94	.94	.95	.94	.94	**.82**
Smiling	.90	.89	.91	.74	.82	.59	.88	**.44**
Straight_Hair	.78	.78	.79	.77	.78	.78	.78	**.49**
Wavy_Hair	.78	.76	.77	.77	.78	.77	.78	**.52**
Wearing_Earrings	.90	.89	.86	.88	.85	.89	.88	**.61**
Wearing_Hat	1.00	1.00	.97	.97	.97	1.00	.97	**.94**
Wearing_Lipstick	.90	.90	.90	.89	.85	.88	.92	**.75**
Wearing_Necklace	.84	.85	.80	.80	.77	.85	.83	**.68**
Wearing_Necktie	.93	.93	.92	.92	.91	.92	.93	**.78**
Young	.86	.86	.85	.84	.83	.86	.87	**.60**
Mean	.88	.88	.87	.86	.84	.86	.87	**.71**

Justification: **[TODO]**

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Table 14: Per-concept mean control accuracy change ΔCtrl (%) on CelebA, all 40 binary attributes. For each concept, ΔCtrl is averaged over the 5 least-correlated control attributes (closer to zero = more surgical). λ^* selection matches Table 13.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	+0.0	+0.1	-0.7	+0.1	+0.0	+0.1	-2.3	-0.2
Arched_Eyebrows	-1.3	-2.3	-1.2	-0.8	-1.1	-2.5	+0.6	-0.6
Attractive	-0.3	-1.0	-1.7	-0.8	-0.3	-0.9	+0.0	-0.8
Bags_Under_Eyes	-3.7	-1.7	-0.3	-7.7	-0.3	-0.3	-0.3	-0.1
Bald	+0.1	+0.0	-0.1	+0.1	-0.1	+1.1	+2.3	+0.2
Bangs	-3.7	-1.2	-0.5	-7.9	-0.1	-0.7	-0.3	-0.8
Big_Lips	+0.1	-0.0	-0.3	+0.0	-0.1	-0.5	+0.1	-0.1
Big_Nose	-0.1	+0.3	-3.5	-9.7	+0.0	-0.3	-1.8	+0.1
Black_Hair	-0.1	-0.0	+0.1	+0.2	+0.3	-0.3	-0.3	-0.7
Blond_Hair	-0.3	+0.1	-1.4	+0.7	-0.1	-0.4	-1.1	-1.6
Blurry	-1.3	+0.0	-0.1	-5.5	+0.1	-0.7	+0.7	-0.1
Brown_Hair	+0.1	-0.9	-0.3	-6.0	-0.7	+0.4	+0.1	-0.2
Bushy_Eyebrows	+0.0	+0.1	-0.1	-1.5	+0.2	-0.9	-0.3	-0.5
Chubby	-0.1	+0.3	-0.2	-1.4	-0.2	-0.2	+0.4	-2.0
Double_Chin	+0.1	+0.1	-0.4	-12.7	+0.3	+0.7	-0.7	-2.5
Eyeglasses	-0.1	+0.1	-0.0	-0.1	-0.2	-0.4	-0.2	+0.2
Goatee	+0.1	-0.1	+0.0	+0.2	+0.1	+0.5	-0.3	-0.4
Gray_Hair	-0.2	-0.3	-2.1	+0.4	-0.1	-1.1	-1.5	-1.5
Heavy_Makeup	-0.3	-0.1	-1.1	-1.7	-1.3	-1.3	-1.1	-0.9
High_Cheekbones	+0.1	+0.2	+0.1	-0.3	-0.1	-0.4	-0.0	-0.7
Male	-0.3	-0.1	-0.7	-0.2	-0.3	-2.2	-0.9	-1.1
Mouth_Slightly_Open	-0.3	-0.6	+0.1	+0.4	+0.2	+0.3	+0.1	-0.2
Mustache	+0.0	-0.1	+0.1	-10.3	-0.5	+1.0	+0.5	-1.5
Narrow_Eyes	-0.1	-0.2	-0.3	+0.2	-0.1	+0.3	+0.0	+0.0
No_Beard	-1.2	-0.2	-4.0	-6.5	-1.6	-0.1	-0.7	-0.5
Oval_Face	-2.0	+0.1	-0.1	-14.4	-0.6	-0.3	+0.2	-0.3
Pale_Skin	-1.2	-0.1	-0.1	-6.8	-0.9	-0.1	+0.7	-1.0
Pointy_Nose	-0.1	-0.2	+0.3	-1.1	+0.2	-0.7	-0.1	-0.1
Receding_Hairline	-0.1	-0.1	-0.1	-5.9	+0.0	-0.3	+0.1	-1.1
Rosy_Cheeks	-0.5	+0.1	-0.6	-0.1	-0.2	-0.1	-0.1	-1.3
Sideburns	-0.4	+0.0	-0.1	+0.1	+0.1	-0.6	-2.5	-0.1
Smiling	-0.3	-1.3	+0.1	+0.6	+0.4	+0.1	+0.1	-0.1
Straight_Hair	-0.1	-0.2	-0.1	-0.9	-0.1	-0.8	+0.5	-0.7
Wavy_Hair	+0.1	-0.7	-0.9	+0.2	+0.1	-0.4	-0.1	-0.7
Wearing_Earrings	+0.8	-0.3	-1.6	-0.3	-0.1	-1.3	-0.5	-0.3
Wearing_Hat	-0.1	+0.0	-0.1	-0.6	+0.0	+1.0	+0.1	-0.2
Wearing_Lipstick	+0.2	-0.1	-6.1	-21.5	-1.9	-1.2	-1.4	-0.9
Wearing_Necklace	+0.1	-1.7	-1.7	-9.3	+0.3	-1.8	+0.2	-1.2
Wearing_Necktie	+0.0	+0.1	+0.0	+0.2	-0.2	-0.7	-13.4	-1.3
Young	-0.0	-0.7	-1.7	-10.7	+0.1	-0.9	-2.2	-0.8
Mean	-0.4	-0.3	-0.8	-3.5	-0.2	-0.4	-0.6	-0.7

Table 15: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: Q1 examines whether task-relevant computation concentrates in a low-dimensional linear subspace, Q2 tests whether perturbations on the manifold are more consequential than off-manifold perturbations, Q3 validates whether the manifold is connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Q1: Rank-retention	Subspace concentration	Top-1 agreement under intrinsic projection
Q2: Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Q3: Domain connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers